TECHNICAL SPECIFICATION

Boat Quotation SaaS — V1

Multi-Tenant Platform for Boat Dealerships

Version	V3.0
Date	April 2026
Status	Reference document for the development team
Architecture	Two-tier SaaS — Platform owner + Dealership tenants
Recommended stack	Next.js · Prisma · PostgreSQL · Supabase · Resend
Build method	Code generation via Claude AI + developer integration

This document is the complete reference for V1 development. The platform operates on two tiers: a global catalogue managed by the platform owner, and isolated workspaces for each dealership tenant. Code will be generated by Claude AI — the developer's role is to integrate, test, and deploy.

1. Product Vision

This is a multi-tenant SaaS platform that enables boat dealerships to create, manage, and send professional quotations and order confirmations in under 2 minutes.

The platform is designed to be sold as a subscription to multiple dealerships. Each dealership gets its own isolated workspace. The platform owner maintains a shared global catalogue of boat brands, models, and options that any dealership can activate and use immediately.

> *This is NOT a management software. It is a sales acceleration tool — built to scale across many dealerships.*

Two-Tier Architecture Overview

Tier	Who	What they manage
Platform (Superadmin)	The software owner	Global brands, models, variants, options, pricing — the shared catalogue
Dealership (Tenant)	Each subscribing dealership	Their own clients, quotes, settings, private brands — fully isolated

Key Principles

- A dealership activates brands from the global catalogue — no manual data entry needed to get started
- When the platform owner updates a brand's catalogue, subscribed dealerships are notified and can choose to pull the update
- Dealerships can also add their own private brands and models not in the global catalogue
- All quotes are snapshots — catalogue updates never break existing quotes
- Each dealership's data is strictly isolated — no cross-tenant data access is ever possible

Success Criteria

- A new dealership can onboard and create their first quote in under 10 minutes
- A complete quote can be created in under 2 minutes
- No repetitive manual input is required
- The interface is fluid and updates instantly without page reloads
- The generated PDF is clean and professional
- Email sending works reliably with PDF attachment
- Order confirmation (bon de commande) can be generated in one click from a won quote

Out of Scope — V1

- Accounting, invoicing, inventory management
- Workshop management, marketplace, listing distribution
- Native mobile application
- Brands managing their own catalogue directly (platform owner manages all global data in V1)

2. Technical Architecture

Recommended Stack

Layer	Technology	Reason
Frontend	Next.js + Tailwind CSS	SSR, routing, optimal performance
State management	Zustand	Live quote calculations without page reloads
Backend / API	Next.js API Routes	Simplifies full-stack deployment
ORM	Prisma	Typed schema, clean migrations
Database	PostgreSQL (Supabase)	Relational, robust, multi-tenant ready
Auth	Clerk or NextAuth	Multi-tenant, role-based access (superadmin vs tenant)
PDF generation	Puppeteer or @react-pdf	Server-side PDF rendering
Email	Resend	Simple API, attachment support, reliable delivery
Excel/CSV import	SheetJS (xlsx)	Catalogue file parsing
Hosting	Vercel + Supabase	Simple deployment, scalable infrastructure

User Roles

Role	Access level	Description
Superadmin	Platform-wide	Platform owner — manages global catalogue, all tenants, and platform settings. Accessed via a separate /admin route.
Tenant admin	Company-scoped	Dealership owner or manager — manages their company settings, activates brands, manages their team.
Tenant user	Company-scoped	Salesperson — creates and manages quotes within their dealership workspace.

Authentication & Isolation

- Email + password login for all roles
- Password reset via email
- Superadmin access is role-gated — no dealership user can access /admin routes
- All tenant queries are scoped by company_id — enforced at the database query level
- No company's data can ever be accessed by another company or by a tenant user

3. Database Schema

The schema is split into two tiers: platform-level tables (global catalogue) and tenant-level tables (per-dealership data). All tenant tables include company_id for strict isolation.

Platform-Level Tables (Global Catalogue)

Table	Role
BRAND	Global boat brands maintained by the platform owner (e.g. Brig, Jeanneau, Quicksilver)
GLOBAL_BOAT_MODEL	Global models per brand with unique code, name, and default margin
GLOBAL_BOAT_VARIANT	Global variants per model with base price, cost, and currency
GLOBAL_EQUIPMENT_ITEM	Global reusable equipment library
GLOBAL_VARIANT_EQUIPMENT	Equipment included in a global variant (standard or free text)
GLOBAL_OPTION	Global options per model with category, price, cost, and position
CATALOGUE_UPDATE_LOG	Log of every change made to the global catalogue by the superadmin

Tenant-Level Tables (Per Dealership)

Table	Role
COMPANY	Dealership profile, legal details, settings, logo, salesperson
USER	Users belonging to a company with role (admin / user)
COMPANY_BRAND	Brands activated by the dealership from the global catalogue
MARGIN_PRESET	Margin rates per category (hull, engine, options, custom items)
EMAIL_TEMPLATE	Editable email templates per document type
CLIENT	Customer contacts with internal notes
COMPANY_BOAT_MODEL	Boat models in the dealership workspace — either copied from global or created privately
COMPANY_BOAT_VARIANT	Variants in the dealership workspace

COMPANY_EQUIPMENT_ITEM	Equipment library in the dealership workspace
COMPANY_VARIANT_EQUIPMENT	Equipment included in a company variant
COMPANY_OPTION	Options available in the dealership workspace
IMPORT_LOG	History of catalogue imports (file, rows, errors)
DOCUMENT	Brochures attached to a model or variant
CATALOGUE_UPDATE_NOTIFICATION	Pending update notifications from the global catalogue for this dealership
QUOTE	Full quote with financial snapshot, status, trade-in details
QUOTE_OPTION	Options selected in a quote (price/cost/discount snapshot)
QUOTE_CUSTOM_ITEM	Manual line items (transport, preparation, admin fees...)
ORDER_CONFIRMATION	Bon de commande generated from a won quote
EMAIL_LOG	Email send history per quote or order confirmation

Brand Activation Model

When a dealership activates a brand from the global catalogue, the system copies the full catalogue (models, variants, equipment, options) into their company workspace. This gives them:

- A working catalogue from day one — no manual data entry
- The ability to customise prices and margins independently from the global catalogue
- Protection from automatic global updates breaking their existing configuration

Copy-on-activation approach: the global catalogue is the source of truth for the platform owner. Each dealership holds their own copy, which they can customise. Updates from the global catalogue are pushed as notifications — the dealer decides when and whether to apply them.

Margin Resolution Logic

When no cost price is provided, the system estimates the margin using the following cascade in strict priority order:

Priority	Source	Condition

1 — Highest	Real margin (calculated)	Cost price has been entered for this item
2	MARGIN_PRESET by category	A preset exists for this category (hull, options...)
3	COMPANY_BOAT_MODEL.default_margin_pct	A margin has been set on this specific model
4 — Fallback	COMPANY.default_margin_pct	Global company default

Every quote line clearly indicates whether the margin shown is REAL (cost provided) or ESTIMATED (calculated via cascade). This is visible in the quote builder but never shown in the client PDF.

4. Superadmin Panel

The superadmin panel is a separate area of the application accessible only to the platform owner. It is completely hidden from all dealership users. It is accessed via a dedicated route (e.g. /admin) and requires superadmin authentication.

4.1 Global Catalogue Management

The superadmin manages the full global catalogue. All changes are versioned in CATALOGUE_UPDATE_LOG and can trigger notifications to subscribed dealerships.

Section	What the superadmin can do
Brands	Create, edit, deactivate brands. Set brand logo, description, display order.
Global models	Create, edit, archive models. Set unique code, name, brand, default margin.
Global variants	Create, edit variants per model. Set base price, cost, currency.
Global equipment	Manage the global equipment library used as standard features in variants.
Global options	Create, edit, archive options per model. Set category, price, cost, position.
Bulk import	Import or update the global catalogue via Excel/CSV for any brand.

4.2 Catalogue Update Notifications

When the superadmin modifies any global catalogue item (price change, new model, new option, archived variant), the system automatically generates a CATALOGUE_UPDATE_NOTIFICATION for every dealership that has activated that brand.

Notification field	Description
brand	Which brand was updated
change_type	new_model / price_update / new_option / variant_archived / option_archived
summary	Human-readable description of what changed (e.g. Seagame 250 — 3 new options added)
affected_items	List of specific models, variants, or options that changed
created_at	Date and time of the change

status	pending / applied / dismissed per dealership

4.3 Dealership Management

Action	Description
View all tenants	List of all subscribed dealerships with status, activation date, active brands
Create a dealership	Manually onboard a new dealership and set their initial brand access
Suspend / reactivate	Temporarily block access to a dealership workspace
View usage stats	Number of quotes created, last activity date per dealership

4.4 Superadmin Dashboard

- Total number of active dealerships
- Total quotes created across all tenants this month
- Most activated brands across all dealerships
- Recent catalogue update activity

5. Dealership Onboarding Flow

When a new dealership subscribes, they go through a guided onboarding flow before accessing the main application.

Step	Action	Details
1	Create account	Company name, email, password
2	Company profile	Legal details: name, legal form, SIREN, VAT number, address
3	Upload logo	Company logo for PDF headers
4	Salesperson details	Name, phone, email
5	Activate brands	Browse the list of available global brands and activate the ones they sell
6	Catalogue copied	System copies all models, variants, and options for activated brands into their workspace
7	Ready	Dealership lands on their dashboard — catalogue is ready, first quote can be created immediately

Goal: a new dealership can go from sign-up to creating their first quote in under 10 minutes, with zero manual catalogue entry for global brands.

6. Brand Activation & Catalogue Updates

6.1 Activating a Brand

From their settings area, a dealership can browse the list of all available global brands and activate the ones they sell. They can also deactivate brands they no longer carry.

- Activated brands appear in their catalogue immediately
- All models, variants, equipment, and options are copied into their workspace
- The dealership can customise prices and margins on their copy independently

6.2 Adding a Private Brand

If a dealership sells a brand not available in the global catalogue, they can create it manually in their workspace. Private brands and their models are fully owned by the dealership and are never shared with other tenants or visible in the global catalogue.

- Same data structure as global brands: models, variants, equipment, options
- Fully private — not visible to other dealerships or to the superadmin catalogue
- Can be imported via Excel/CSV using the same import template

6.3 Receiving Catalogue Updates

When the platform owner updates a global brand's catalogue, all dealerships with that brand activated receive a notification in their workspace. The notification appears as a banner or alert in their dashboard.

Action available	Description
Apply update	The new models, variants, or prices are merged into the dealership's workspace copy. Existing quotes are never affected.
Review changes	The dealership can view exactly what changed before deciding to apply
Dismiss	The dealership ignores the update and keeps their current catalogue version

Existing quotes are always protected. A catalogue update only affects new quotes created after the update is applied. All financial data in existing quotes is stored as a snapshot at creation time.

7. Catalogue Management — Dealership

7.1 Models and Variants

Each model in the dealership workspace has a unique code (e.g. SG250). Models can come from the global catalogue (copied on brand activation) or be created manually as private models.

Field	Type	Description
code	unique string	Business identifier e.g. SG250 — used for import matching
name	string	Commercial model name
brand	string	Brand name
source	enum	global (copied from platform) or private (created by dealership)
default_margin_pct	float	Default margin — overrides company default for this model

7.2 Included Equipment

Each variant can include equipment of two types:

- Standard — selected from the equipment library
- Free text — manually typed (preparation included, special configuration, custom notes)

7.3 Options

Options are paid items linked to a specific model, organised by categories. They appear in the quote builder as checkboxes with automatic price display.

7.4 Catalogue Import (Private Brands)

Recommended import template

Column	Required	Description
model_code	Yes	Target model code e.g. SG250
category	Yes	Option category e.g. CC Configuration
label	Yes	Option name / description

price	Yes	Selling price excl. VAT
cost	No	Cost price (optional)
position	No	Display order within the category

3 supported import methods

- **Method 1:** File upload — Excel (.xlsx) or CSV
- **Method 2:** Copy-paste — paste directly from Excel, tab-separated columns detected automatically
- **Method 3:** Manual entry — option by option via the catalogue UI

Import behaviour

- Auto-detection of category header rows — any row without a price is treated as a category name
- Creates new entries that do not exist yet
- Updates existing entries matched on model_code + label
- Detailed error report on completion
- Full import history in IMPORT_LOG

8. Quote Builder

8.1 User Flow

The quote builder is the core of the product. It must be fast, fluid, and update without any page reloads. It works identically whether the model comes from the global catalogue or is a private brand.

Step	Action	Expected behaviour
1	Select or create a client	Instant search, inline quick-create form
2	Select a boat model	List filterable by brand and model — shows all activated brands + private brands
3	Select a variant	Automatically loads included equipment and available options
4	Add options	Checkboxes with prices — live total update
5	Add custom line items	Transport, preparation, fees — free text + price
6	Apply discounts	Per item, per category, or global discount
7	Add a trade-in	Trade-in form with automatic deduction from total
8	Review the summary	Right-side live panel with all financial totals
9	Generate the quote	PDF generated + email sending option

8.2 Interface Layout

- Two-column layout: left = configuration panel, right = live financial summary
- Instant updates with no page reloads — state managed via Zustand
- HT or TTC display mode — configurable per quote, defaults to company setting

8.3 Live Financial Summary

- Base price (excl. VAT)
- Options total
- Applied discounts broken down by level
- Total HT, VAT amount, Total TTC

- Trade-in deduction (if added)
- Net amount payable
- Total cost (if provided)
- Margin in euros and percentage — with REAL / ESTIMATED indicator

9. Discounts

Level	Scope	Example
Item	One specific option or line item	10% off the solar panel kit
Category	All options within a category	5% off the CC Configuration category
Global	The entire quote	3% global commercial discount

Calculation order: item discount first, then category discount, then global discount. All discounts are shown in the PDF and the live summary.

10. Trade-In

The trade-in value is deducted from the total TTC and displayed clearly in the PDF and order confirmation.

Field	Type	Required
Brand	string	No
Model	string	No
Year	string	No
Engine	string	No
Engine hours	integer	No
Description	free text	No
Trade-in value	float	Yes

11. Quote Lifecycle & Management

11.1 Automatic Numbering

Quotes: Q-YYYY-NNN (e.g. Q-2026-001). Order confirmations: BC-YYYY-NNN (e.g. BC-2026-001). Both counters are per-dealership and reset each year.

11.2 Quote Statuses

Status	Description	Next action
Draft	In progress — not yet sent	Send by email
Sent	PDF transmitted to client	Follow-up, mark Won or Lost
Won	Order confirmed	Generate order confirmation PDF
Lost	Deal not closed	Duplicate and restart if needed

11.3 Quote Duplication

Any quote can be duplicated in one click — created as Draft with a new number. Essential for negotiation workflows with multiple versions of the same quote.

11.4 Internal Notes

Free-text internal notes on every quote and every client. Never visible in any PDF or email.

12. PDF Generation — Quote

PDF Section	Content
Header	Company logo, salesperson name, phone, email
Legal footer	Company name, legal form, SIREN, VAT number, address
Client details	Full name, email, phone, address
Quote reference	Quote number, creation date
Boat	Model name, variant, optional specifications
Included equipment	Standard features — visually separated from options
Selected options	Paid options with unit price and discounts
Custom line items	Transport, preparation, admin fees etc.
Discounts	Breakdown by level
Trade-in	Details and deducted value (if applicable)
Totals	Total HT, VAT, Total TTC, Trade-in, Net payable

Margin, cost prices, and internal notes must never appear in the client PDF under any circumstance.

13. Order Confirmation (Bon de Commande)

When a quote is marked Won, the user can generate a separate order confirmation PDF. This is a distinct document from the quote — it is a formal commitment and cannot be edited after generation.

PDF Section	Content
Header	Company logo, salesperson name, phone, email
Document title	ORDER CONFIRMATION / BON DE COMMANDE
Reference	Order number (BC-YYYY-NNN), date, linked quote number
Client details	Full name, email, phone, address
Boat	Model, variant, full configuration summary
Included equipment	Standard features
Selected options	All paid options with final agreed prices
Custom line items	Transport, preparation, fees etc.
Final totals	Total HT, VAT, Total TTC, Trade-in, Net payable
Signature block	Client signature + salesperson signature + date
Legal mentions	SIREN, VAT number, legal form, address, payment terms

14. Email Sending & Templates

14.1 How It Works

The user sends quotes and order confirmations directly from the application. The correct PDF is automatically attached. Email content is pre-filled from the configured template and is fully editable before sending.

Template	Triggered when	Attached document
Quote sending	User clicks Send on a quote	Quote PDF
Order confirmation	User sends a bon de commande	Order confirmation PDF
Follow-up reminder	User manually triggers a follow-up on a Sent quote	Quote PDF (resent)

14.2 Dynamic Variables

Variable	Replaced with
{{client_name}}	Client full name
{{client_first_name}}	Client first name
{{quote_number}}	Quote reference number
{{order_number}}	Order confirmation number
{{boat_model}}	Boat model and variant name
{{total_ttc}}	Total amount incl. VAT
{{salesperson_name}}	Salesperson name from settings
{{company_name}}	Company name from settings
{{date}}	Today's date

14.3 Template Editor

- Subject line editable with variable support
- Rich text body editable with variable support
- Live preview showing the rendered email with example values
- Reset to default option per template

14.4 Email Tracking

- Send date, recipients, template used, and attached documents stored in EMAIL_LOG
- Quote status automatically updated to Sent on email send

15. Multi-Currency

Field	Description
Cost currency	USD or EUR depending on the supplier
USD/EUR exchange rate	Entered manually by the user
Rate date	Stored as a snapshot with the quote
Display	All amounts shown in EUR in the quote and PDF

Exchange rate is stored as a snapshot per quote. Changes to the rate after quote creation never affect existing quotes.

16. Clients, Quotes & Dashboard

16.1 Clients Page

- Paginated list of all company clients with global search
- Quick access to all quotes for a given client
- Internal notes per client

16.2 Quotes Page

- Full list with status, amount, client, date — filterable and searchable
- Actions: open, edit, duplicate, update status, generate order confirmation

16.3 Dashboard

Indicator	Description
Quotes today	Number of quotes created today
Total value	Total quoted amount this month
Quotes by status	Draft / Sent / Won / Lost counts
Most quoted models	Top 5 models over current period
Pending catalogue updates	Alert if new catalogue updates are available to apply

17. Company Settings

17.1 Company Profile & Legal Details

Setting	Description
Company name	Displayed in all PDFs and emails
Legal form	e.g. SAS, SARL, SA, EI — PDF footer
SIREN number	French business registration number — PDF footer
VAT number	Intracommunity VAT number — PDF footer
Address	Full company address — PDF footer
Logo	Image upload — PDF header

17.2 Salesperson Details

Setting	Description
Salesperson name	PDF header and signature block
Salesperson phone	PDF header
Salesperson email	Email sender address

17.3 Defaults

Setting	Description
Default VAT rate	e.g. 20% — overridable per quote
Default global margin	Fallback if no category or model margin is set
Default display mode	HT or TTC — overridable per quote

17.4 Margin Presets

Category	Description
hull	Margin on the base boat price
engine	Margin on the engine
options	Margin on paid options
custom_items	Margin on manual line items

Full margin priority: Real (cost entered) > Model-level margin > Category preset > Global company default.

17.5 Brand Management

- Browse and activate global brands from the platform catalogue
- View pending catalogue update notifications per brand
- Apply, review, or dismiss catalogue updates
- Create and manage private brands not available in the global catalogue

17.6 Email Templates

Three editable templates: Quote sending, Order confirmation, Follow-up reminder. See Section 14 for full specification.

18. UX & Performance Requirements

18.1 User Experience

- Usable without training — intuitive from the first login
- Modern SaaS design — clean, professional
- Responsive: desktop priority, tablet functional, mobile acceptable
- No page reloads in the quote builder
- Inline form validation with clear error messages

18.2 Performance Targets

Action	Target
Initial page load	Under 2 seconds
Live summary update in quote builder	Under 100ms
PDF generation	Under 5 seconds
Catalogue import (500 rows)	Under 10 seconds
Brand activation (copy catalogue)	Under 15 seconds
Email send with attachment	Under 5 seconds

18.3 Security

- HTTPS enforced on all routes
- Superadmin routes fully separated and role-gated
- Strict data isolation per company_id at query level
- No cross-tenant data access possible under any circumstance

19. Build Method — Claude AI + Developer

Application code will be generated module by module by Claude AI. The developer integrates, connects, tests, and deploys.

The ideal developer profile is a mid-level full-stack developer comfortable with React and Node.js, capable of reading and wiring AI-generated code. Prefer milestone-based contracts.

Phase	Owner	Content
1 — Database schema	Claude AI	Complete Prisma schema — platform + tenant tables
2 — Project setup	Claude AI + Dev	Next.js + Prisma + Supabase + Auth + role system
3 — Superadmin panel	Claude AI	Global catalogue CRUD, tenant management, update notifications
4 — Onboarding flow	Claude AI	Brand activation, catalogue copy, workspace setup
5 — Catalogue module	Claude AI	Tenant catalogue management, private brands, import
6 — Quote builder	Claude AI	Full builder with live calculations
7 — Quote PDF	Claude AI	Professional quote PDF template
8 — Order confirmation PDF	Claude AI	Bon de commande with signature block
9 — Email module	Claude AI	Template engine, variable replacement, sending via Resend
10 — Dashboard & lists	Claude AI	Stats, quote list, client list, update notifications
11 — Company settings	Claude AI	Settings UI, legal details, brand activation, email templates
12 — Integration	Developer	Wiring, testing, edge cases, bug fixes
13 — Production deploy	Developer	Vercel + Supabase + domain + SSL

20. Expected Deliverables

Deliverable	Description	Priority
Deployed application	Public URL accessible from any browser	Critical
Working authentication	Sign up, login, password reset, role system	Critical
Superadmin panel	Global catalogue management + tenant management	Critical
Brand activation flow	Dealership activates brands and receives catalogue copy	Critical
Quote builder	Complete creation flow in under 2 minutes	Critical
Quote PDF	Clean output with legal footer	Critical
Order confirmation PDF	Bon de commande with signature block	Critical
Email sending	Template-based with variable replacement	Critical
Catalogue update notifications	Dealer notified when global catalogue changes	High
Private brand import	Excel / CSV + copy-paste	High
Dashboard	Stats, filterable lists, update alerts	High
Multi-currency USD/EUR	Exchange rate stored per quote	Medium
Versioned source code	Git repository with deployment README	High

— End of Technical Specification V3.0 —